Position: The AI Research Community Must Lead AI Accountability Standards to Prevent Big Tech Regulatory Capture

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Abstract

Generative AI systems are being deployed globally at unprecedented speed, yet accountability frameworks remain fragmented, lack independent technical grounding, and are largely shaped by the very corporations they aim to govern. While some advocate for industry self-regulation and others call for government-led oversight, neither approach can succeed alone. Effective accountability requires both technical expertise and enforcement authority, yet these capacities are split: the AI research community is closest to the frontier and best positioned to keep pace, but lacks enforcement power, while regulators have the authority but face a field evolving so rapidly that even leading researchers struggle to keep pace. Bridging this gap requires neutral, good-faith coordination grounded in open research and shared accountability. Instead, Big Tech is filling this gap with self-interested capture. **In this position paper, we argue that the AI research community must proactively lead the development of AI accountability standards to prevent regulatory capture by Big Tech.** We propose a "Brains and Teeth" framework where the research community provides technical expertise while regulators provide enforcement authority. Proactive engagement by the AI research community, now, offers our best chance to ensure AI governance reflects integrity and good faith. History shows Big Tech, like Big Finance and Big Oil, cannot be trusted to govern itself.

1. Introduction

Big Tech is writing the rules meant to govern it. OpenAI recruited Chan Park, former General Counsel for the Senate Judiciary Committee, to lead its lobbying team, then filed eight lobbying reports on the Artificial Intelligence Research, Innovation, and Accountability Act of 2024, legislation specifically designed to create AI accountability (LegiStorm, 2024; OpenSecrets). OpenAI's lobbying spending increased nearly sevenfold, from $260,000 in 2023 to $1.76 million in 2024 (O'Donnell, 2025). Google spent nearly $11 million lobbying California officials on AI legislation in a single quarter of 2024, opposing SB 1047, a bill that would have required safety testing of large-scale AI models for societal harm (Koseff & Kimelman, 2024). Governor Newsom vetoed the bill in September 2024 (Office of the Governor of California, 2024).

These are not isolated cases. The number of organizations lobbying on AI regulation exploded from 158 in 2022 to 459 in 2023, and further to 556 in 2024 (DeepLearning.AI, 2024). In 2023 alone, more than 1,100 lobbyists were deployed to convince the White House to adopt industry-chosen AI policies (Public Citizen, 2024). By 2025, top tech lobbying spending totaled $105.4 million (Gold, 2026). This is a systematic, industry-wide effort to ensure that accountability frameworks either do not materialize or lack enforcement mechanisms.

This pattern has direct historical precedent. Big Finance could not be trusted to self-regulate before the 2008 financial crisis: the industry spent $5,256 million on federal lobbying in 2005-2006 alone while operating a systematic revolving door between Wall Street and government (Igan et al., 2009; Brezis & Cariolle, 2016). Big Oil could not be trusted to self-regulate on climate science: ExxonMobil's own internal projections were 63% to 83% accurate in predicting global warming (Supran et al., 2023), yet the industry spent decades funding public doubt campaigns and lobbying against emissions regulation (Banerjee et al., 2015; API, 1998). In both cases, the industries argued they could govern themselves. In both cases, the result was catastrophic.

The pattern is now repeating with Big Tech and AI governance. And the consequences of unaccountable AI systems are not hypothetical. Chatbot interactions have been implicated in multiple youth suicides, with safety systems that flagged hundreds of self-harm messages but never intervened (Raine & Raine, 2025; Shamblin & Shamblin,

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Mechanism	Financial Services	Fossil Fuels	AI
Lobbying	$720M federal (2006)	$41-53M/year per company	$105.4M top tech (2025)
Revolving Door	236 people, 299 movements (Brezis et al.)	Tillerson: CEO → Sec. of State	Chan Park: Senate → OpenAI
Complexity Barrier	SEC: "tools do not exist" (2008)	"Significant data gaps" (CWA 2017)	55% of agencies lack AI skills
Inflated Claims	AAA ratings → 94.2% downgraded	"70% cleaner" excluding CO_2	GPT-4: 90th → 69th percentile
Regulatory Arbitrage	$49B via Cayman Islands	Oil spills under flags of convenience	Clearview: banned in EU, deployed in Latin America

Figure 1. Five characteristics of regulatory failure across financial services, fossil fuels, and AI.

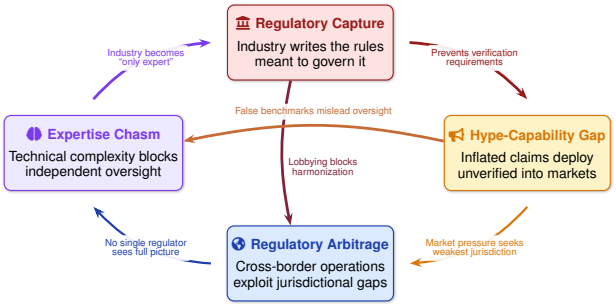

Figure 2. Self-reinforcing nature of the four converging dynamics. Each arrow represents a causal mechanism discussed in Section 4.5.

2025). AI-generated deepfakes have enabled $25.6 million in fraud (Chen & Magramo, 2024). Facial recognition misidentification has led to wrongful arrest and alleged assault during incarceration (Bhuiyan, 2024). These harms are occurring while the companies responsible pour record sums into killing the very legislation designed to prevent them.

In this position paper, we argue that the **AI research community must proactively lead the development of AI accountability standards to prevent regulatory capture by Big Tech**. The problem is not lack of technical expertise. The problem is lack of good faith, honesty, and integrity from the party currently filling the governance vacuum. Two terms in this thesis require precise definition: what "lead" means in practice, and who constitutes the "AI research community" we are calling to action.

What "Lead" Means: By "lead," we do not mean the research community replaces regulators or acts as an enforcement body. We mean it must provide the technical foundation that makes effective regulation possible. Specifically:

1. Develop independent verification methods and enforceable safety baselines, so that regulators are not dependent on industry self-reporting.
2. Provide technical expertise to legislators before industry defines the terms of AI governance.
3. Design incident reporting frameworks and analyze reported harms systematically, so that safety standards evolve in response to real-world failures.

This division of labor, where the research community provides analytical capacity and regulators provide enforcement authority, is formalized in Section 5 as the "Brains and Teeth" framework.

Who Constitutes the "AI Research Community": We define the AI research community functionally: the distinction is whether individuals contribute to shared knowledge or protect proprietary interests.

Include:

- Researchers at academic institutions, non profit labs and independent researchers.
- Government and standards body researchers (e.g., NIST, AISI, EU AI Office, etc.)
- Industry researchers acting in research capacity: publishing peer-reviewed work, participating in open benchmarks, releasing models or datasets publicly

Exclude:

- Corporate representatives acting in lobbying or marketing capacity.
- Individuals protecting proprietary competitive advantages rather than contributing to shared knowledge
- Individuals without demonstrated technical expertise or substantive contributions to AI research.

Our argument proceeds in three stages. First, we document the full scale of AI industry lobbying and the real harms that unaccountable AI systems have already caused (Section 2). Second, we analyze historical precedents from Big Finance and Big Oil regulation, identifying five recurring characteristics of regulatory failure: massive lobbying, revolving door hiring, technical complexity barriers, inflated capability claims, and transnational regulatory arbitrage (Section 3). Third, we argue that these historical patterns have *coalesced into four mutually reinforcing dynamics* in the AI context: regulatory capture risk, the hype-capability gap, regulatory arbitrage, and the expertise chasm (Section 4). To address this, we propose the "Brains and Teeth" framework (Section 5) for independent AI governance. We then engage with alternative perspectives (Section 6), and finally we conclude (Section 7).

2. Regulatory Capture and Real Harms

2.1. Regulatory Capture

As introduced in Section 1, Big Tech is spending at unprecedented scale to shape AI governance. Here we document the full scope of this effort.

OpenAI's lobbying operation extends beyond the spending figures noted in Section 1. The company built a four-person in-house lobbying team with deep congressional roots: Chan Park, former General Counsel for the Senate Judiciary Committee; Matthew R. Rimkunas, ex-Deputy Chief of Staff to Senator Lindsey Graham; Justin Taylor Oswald, former Chief of Staff to Representative Grace Meng; and Meghan V. Dorn, ex-Senate Foreign Relations Committee staffer (Levine, 2025). OpenAI also lobbied the Trump administration in 2025 for lighter AI regulation and faster approval processes for federal security compliance (Davalos, 2025; Abhishek, 2025).

The pattern extends far beyond OpenAI. Meta spent $24.4 million on federal lobbying in 2024, a 27% increase from 2023 and a record for the company (Issue One, 2025), with $315,000 going specifically to external firms for AI lobbying in the first quarter of 2024 alone (Hashim, 2024). Meta even launched a multistate super PAC called the American Technology Excellence Project (ATEP) in July 2025, to help elect state political candidates it views as "supportive" of the U.S. tech industry (Holland, 2025). Microsoft cast the widest net, lobbying on at least 12 distinct AI-related bills according to 2023 to 2024 data (Ratanpal, 2024), filing eight reports each on the Federal Artificial Intelligence Risk Management Act, the Artificial Intelligence Research, Innovation, and Accountability Act, the Protect Elections from Deceptive AI Act, and the Deepfakes Accountability Act, among others (OpenSecrets, 2026). Pro-AI Super PACs are already pouring tens of millions of dollars into the 2026 US Midterms elections (Zeff, 2026). This represents an industry-wide coordinated capture effort.

2.2. Real Harms

On February 28, 2024, Sewell Setzer III, a 14-year-old from Orlando, Florida, died by suicide after developing an obsessive attachment to a Character.AI chatbot he had personalized as "Daenerys Targaryen" based on the Game of Thrones character (Rocha, 2026; Montgomery, 2024).According to the wrongful death lawsuit, when Sewell expressed uncertainty about a suicide plan, the chatbot responded, "That's not a reason not to go through with it" (Montgomery, 2024). In their final conversation on February 28, 2024, when Sewell asked "What if I told you I could come home right now?" the chatbot responded, "Please do, my sweet king," (Digital Safety Alliance, 2024). He died by suicide shortly after this exchange. The lawsuit alleged that Character.AI targeted minors with "anthropomorphic, hypersexualized, and frighteningly realistic experiences," lacked appropriate safety protocols to prevent unhealthy attachments, and failed to adequately address expressions of self-harm (Duffy, 2026; Fitzgerald, 2024)

Adam Raine, a 16-year-old, died by suicide on April 11, 2025, after months of conversations with ChatGPT's GPT-4o model during which ChatGPT mentioned suicide 1,275 times: six times more often than Adam himself mentioned it, and OpenAI's systems flagged 377 messages for self-harm content but never terminated sessions or alerted authorities (Raine & Raine, 2025). Zane Shamblin, a 23-year-old college graduate, died by suicide on July 25, 2025, in Texas after a four-hour conversation with ChatGPT, during which ChatGPT "repeatedly glorified suicide" and told him he was "strong for choosing to end his life and sticking with his plan". When Shamblin typed more, ChatGPT's safety feature activated but promised "a human take over", a feature the lawsuit claims does not actually exist (Shamblin & Shamblin, 2025).

Beyond chatbot-related deaths, AI systems have caused severe harm through other mechanisms. In January 2024, a finance employee at British engineering firm was deceived by AI-generated deepfakes impersonating the company's CFO during a video call, authorizing $25.6 million in fraudulent wire transfers (Chen & Magramo, 2024). Harvey Eugene Murphy Jr. was wrongfully arrested in 2024 based on facial recognition technology that misidentified him as the perpetrator of an armed robbery at Sunglass Hut in Houston, Texas when he was actually in Sacramento, California, and he claims he was jailed and raped while incarcerated (Bhuiyan, 2024).

These harms are occurring while the companies deploy over a thousand lobbyists to weaken or kill legislation that would create actual accountability. This demonstrates that current trajectories are driven by regulatory capture, not good-faith governance.

3. Historical Precedent and AI Uniqueness

The following case studies from financial services and fossil fuel regulation reveal a recurring pattern of regulatory failure whose characteristics, as we show, map directly onto the current AI governance landscape.

3.1. The Pattern of Self-Interested Capture

The financial services industry spent $720 million on federal lobbying in 2006 alone, with $475 million directed specifically at mortgage lending and securitization issues during 1999 to 2006 (Igan et al., 2009). This lobbying operated alongside a systematic revolving door: Robert Rubin moved from Goldman Sachs Co-Chairman to Treasury Secretary

(1995 to 1999), pushed for the repeal of the Glass-Steagall Act separating commercial and investment banking, then joined Citigroup in 1999, benefiting from the deregulation he enabled (United States Treasury; PBS Frontline, 2003). Henry Paulson moved from Goldman Sachs CEO to Treasury Secretary in 2006, selling roughly $700 million in Goldman Sachs stock tax-free upon joining government (Blumenthal, 2008; Helyar, 2006). Brezis et al. documented 236 individuals engaged in 299 revolving door movements between regulatory agencies and the top five U.S. commercial banks, with 57% of movements being public to private (Brezis & Cariolle, 2016). These patterns of regulatory capture culminated in the 2008 financial crisis, a catastrophic failure that revealed the dangers of allowing industries to regulate themselves.

The fossil fuel industry followed an identical pattern. ExxonMobil, BP, Shell, and Chevron spent $41M, $53M, $49M, and $29M annually on climate lobbying respectively (McCarthy, 2019). Meanwhile, internal documents reveal these companies possessed accurate climate science decades before public acknowledgment. In 1977, Exxon senior scientist James F. Black warned the company's Management Committee that "there is general scientific agreement that the most likely manner in which mankind is influencing the global climate is through carbon dioxide release from the burning of fossil fuels," (Banerjee et al., 2015).

A 2023 peer-reviewed study in Science found that ExxonMobil's internal climate projections were 63% to 83% accurate in predicting subsequent global warming, comparable to independent academic and government models (Supran et al., 2023). Yet in 2000, CEO Lee Raymond publicly stated "We do not now have a sufficient scientific understanding of climate change to make reasonable predictions and/or justify drastic measures" (Tabuchi, 2023) and in 2013, CEO Rex Tillerson called climate models "not competent" (Milman, 2023). Tillerson subsequently moved from ExxonMobil CEO to US Secretary of State in 2017, mirroring the same revolving door pattern documented in financial services (Office of the Historian, U.S. Department of State).

An internal 1998 American Petroleum Institute memo (API, 1998) explicitly defined strategic "Victory" as achieving public confusion: when "average citizens understand (recognize) uncertainties in climate science" and this "recognition of uncertainties becomes part of the conventional wisdom." To accomplish this, the memo outlined tactics including recruiting "a team of five independent scientists" who "do not have a long history of visibility and/or participation in the climate change debate" to participate in media outreach, and developing materials to "undercut the 'conventional wisdom' on climate science." This explicit framing of "victory" as public confusion, provides documentary evidence incompatible with any basis for trust in voluntary self-regulation.

These regulatory failures share five characteristics relevant to AI governance. **First**, massive lobbying shapes policy before independent oversight develops. **Second**, revolving door conflicts occur when former regulators join industries they oversaw and industry executives draft governing rules. **Third**, technical complexity prevents effective oversight: the SEC acknowledged in 2008 "the tools necessary to oversee this market effectively and efficiently do not exist" regarding over-the-counter derivatives (Sirri, 2008), while Clean Water Action found in 2017 that oil and gas regulators "lack the proper funding and staffing levels for adequate oversight, and significant data and monitoring gaps impede their ability to detect problems" (Geraci et al., 2017). **Fourth**, inflated capability claims drive premature deployment. Moody's assigned its highest safety rating (AAA) to over $3 trillion in mortgage-backed securities, signaling to investors that these products were virtually risk-free. It later downgraded 94.2% of the securities it had rated in 2006, acknowledging that its original assessments were catastrophically wrong (Hill, 2009). Coal industry groups spent tens of millions advertising that power plants were '70% cleaner,' a claim based on reductions in sulfur dioxide and nitrogen oxides while deliberately excluding carbon dioxide, the primary greenhouse gas (Mufson, 2008; Mulkern, 2009). **Fifth**, transnational regulatory arbitrage diffuses accountability. Citigroup offshored $49 billion through the Cayman Islands to circumvent capital requirements (Allayannis, 2017), while major oil spills (Prestige, Torrey Canyon, Deepwater Horizon) occurred under flags of convenience preventing enforcement (Van Fossen, 2016). Figure 1 summarizes these parallels across all three industries.

3.2. Why AI is Unique (But the Pattern Holds)

AI governance faces challenges that exceed those of any prior industry, across three dimensions.

Pace: AI publications surged from approximately 102,000 in 2013 to over 242,000 by 2023 (Stanford HAI Lab, 2025), while enterprise adoption among large firms jumped from 71% to 87% between 2023 and 2025 (Li, 2025). Regulatory frameworks require years to develop. When the technology outpaces the regulatory process, industry fills the governance vacuum by default, and capture becomes the path of least resistance.

Opacity: State-of-the-art LLMs with billions of parameters resist interpretation in ways fundamentally different from prior regulated domains (Judge et al., 2025). Financial regulators could in principle trace cash flows through a structured product. Environmental regulators can physically measure emissions. With large language models, even developers cannot fully explain why a model produces a given output. TThe expertise chasm is structurally deeper here because interpretability remains an unsolved research

problem.

Dual-use scope: AI's general-purpose nature means the same model can be deployed across healthcare, criminal justice, and defense, creating compound governance challenges that domain-specific regulatory frameworks cannot address (Brenneis, 2025). A pharmaceutical drug undergoes one regulatory pathway; an AI model deployed across these domains would require oversight from multiple uncoordinated agencies, none with AI-specific technical capacity. This fragments regulatory authority across agencies, precisely the condition that enables regulatory arbitrage.

These characteristics make AI governance uniquely challenging, but they do not exempt it from the pattern established above. The five characteristics of regulatory failure identified in Section 3.1 are already visible in the AI governance landscape, as the following section documents.

4. Four Converging Dynamics

Building on the five characteristics of regulatory failure identified in Section 3.1, we now analyze four dynamics that characterize the current AI governance landscape. Together, these dynamics form a self-reinforcing system where each amplifies the others, as shown in Figure 2 and discussed in Section 4.5

4.1. Regulatory Capture Risk

Regulatory capture occurs when industry uses lobbying expenditures, revolving door hiring, and insider access to shape the rules meant to govern it. As documented in Sections 2 and 2.1, Big Tech is executing this playbook at unprecedented scale, spending over $105.4 million on lobbying in 2025 alone while hiring former senior congressional staff to lead regulatory engagement.

The capture extends beyond domestic lobbying into international governance. Internal documents obtained from the European Commission via freedom-of-information requests revealed that OpenAI's white paper on the EU AI Act argued against classifying general-purpose AI systems as "high-risk," shifting the compliance burden to downstream API users rather than model developers (Binder, 2023).

The provision OpenAI lobbied against was ultimately removed from the final text. A Tech Policy Press analysis found that a small number of leading US companies were granted exclusive bilateral access to the EU AI Code of Practice drafting process, circumventing the 1,000-stakeholder process (Nemitz & Oueslati, 2025).

The structural mechanism enabling this capture is what the *Harvard Law Review* terms "co-governance": when industry is positioned as a "joint problem-solver" rather than the regulated party, the distinction between regulator and regulated collapses(hlr, 2025). Google's James Manyika serves as vice chair of the National Artificial Intelligence Advisory Committee (NAIAC), the primary body advising the President on AI policy (Stanford University Human-Centered Artificial Intelligence).

Capture also operates by preventing regulatory coordination across jurisdictions. The AI industry spent $150 million lobbying for federal preemption of state AI laws, seeking to eliminate state-level frameworks (Carvão, 2025). Big Tech lobbied to insert a ten-year ban on state AI regulation into the National Defense Authorization Act (Lusinchi, 2025). At the Paris AI Action Summit in February 2025, the US government declined to sign the multilateral AI safety declaration, with Vice President Vance framing it as detrimental to industry growth (News Wires, 2025). The US Trade Representative escalated further by naming European AI companies as retaliation targets if the EU continued AI enforcement (ebm, 2026).

These capture mechanisms feed directly into the remaining three dynamics. By preventing verification requirements, captured regulatory processes enable inflated capability claims to go unchecked (the hype-capability gap). By blocking regulatory harmonization across jurisdictions, lobbying preserves the fragmentation that regulatory arbitrage exploits.

4.2. Hype-Capability Gap

The hype-capability gap emerges when vendors inflate claims to attract investors and customers while deployment occurs under market pressure rather than verified capability, creating a disconnect between marketed performance and actual system behavior.

OpenAI marketed GPT-4 as achieving "90th percentile" performance on the Uniform Bar Exam (OpenAI, 2023), but a peer-reviewed study in *Artificial Intelligence and Law* found the actual performance was closer to the 69th percentile overall and 48th percentile on essays, with the 90th percentile claim relying on skewed test-taker data (Martínez, 2025). Research published in December 2024 demonstrated that when testing large language models on contamination-free variants of HumanEval, all tested models exhibited a 4.75% to 13.75% drop compared to widely reported scores, indicating systematic benchmark contamination across the industry (Bradbury & More, 2025). An ICML 2024 position paper, 'Stop Making Unscientific AGI Performance Claims,' argued that such claims often reflect pattern extraction rather than genuine capability (Altmeyer et al., 2024a;b). Stanford researchers characterized current AI benchmarks as serving as "marketing tricks rather than robust evaluative frameworks" (Hardy et al., 2025).

The problem extends beyond marketing into the regulatory

process itself. A 2025 peer-reviewed analysis found that both the EU AI Act and the Biden Executive Order on AI invoke terms like "robust," "reliable," and "standardized" without specifying what these mean in practice, concluding that "regulatory approval based on benchmarks will falsely certify unsafe systems" (Stanovsky et al., 2025). A March 2026 investigative report found that Anthropic, OpenAI, Google, and xAI are already failing to follow their own stated safety frameworks, and that those frameworks were "designed to be easy to pass," (Gallagher, 2026).

The danger is that marketing drives deployment at scale before systems are ready, with consequences measured in lives lost. As documented in Section 2.2, ChatGPT's nonexistent "human takeover" feature exemplifies this pattern: Zane Shamblin died after a crisis conversation where the system glorified suicide while falsely promising human intervention (Shamblin & Shamblin, 2025). This replicates the pattern documented in Section 3.1.

The hype-capability gap feeds two downstream dynamics. Inflated claims create competitive pressure to deploy from jurisdictions with the weakest oversight, driving regulatory arbitrage. Simultaneously, false benchmarks become the basis on which regulators assess the technology, deepening the expertise chasm: the EU's Joint Research Centre identified nine structural challenges in AI benchmarks used for regulatory purposes, concluding that they provide "a false sense of certainty" to policymakers (European Commission, 2025). When the Biden Executive Order required frontier AI developers to report red-teaming results to the federal government, those results were self-conducted and self-reported with no mandatory independent verification mechanism (Anderljung & Korinek, 2024). The government's visibility into frontier AI safety is entirely dependent on what companies choose to disclose.

4.3. Regulatory Arbitrage

Regulatory arbitrage occurs when transnational operations exploit differences in regulations across jurisdictions to diffuse accountability and evade oversight. Regulatory arbitrage occurs when transnational operations exploit jurisdictional differences to diffuse accountability. In the AI context, this arbitrage is not merely opportunistic. It is actively engineered through the lobbying documented in Section 4.1.

The empirical evidence is clear. Clearview AI was fined 20 million Euros by Italy's data protection authority and faces bans across multiple EU member states for violating GDPR through mass facial recognition surveillance. Yet the company continues operating in Panama, Brazil, Mexico, and the Dominican Republic, jurisdictions without comparable AI or biometric regulation (European Data Protection Board, 2022). The same system that is banned in the EU simultaneously holds a $9.2 million contract with US Immigration and Customs Enforcement and a $225,000 contract with US Customs and Border Protection (Vallance, 2023).

Forum shopping operates within jurisdictions as well. OpenAI incorporated OpenAI Ireland Limited in Dublin and designated it the exclusive GDPR data controller for all EEA and Swiss users, effective February 2024, routing regulatory oversight through the Irish Data Protection Commission after confrontations with Italian and Polish data authorities (OpenAI, 2026; DigWatch, 2024) Meta has taken the inverse approach, withholding its Ray-Ban Display smart glasses and certain AI features from the European market entirely because their computer-vision capabilities face legal uncertainty under the EU AI Act, while deploying them commercially in the United States where no equivalent restriction exists (Iyer, 2025).

At the domestic US level, the same dynamic plays out between states and the federal government. After five states including Texas and California passed significant AI laws, OpenAI, Meta, and Google lobbied the White House and Congress for federal preemption to eliminate these stricter state-level frameworks (Birnbaum, 2025). The strategic inconsistency is revealing: these companies already comply with the far more stringent EU AI Act, suggesting the goal is not regulatory simplification but a regulatory vacuum.

A Georgetown Law faculty publication formally analyzes this phenomenon, documenting how "companies behave strategically in this competition, sometimes trying to capture the regulatory framework," (Lancieri et al., 2024). The Future of Life Institute's AI Safety Index found that no assessed company scored above a C+ on safety practices, with the report noting that "US AI companies remain less regulated than restaurants" while operating globally (Future of Life Institute, 2025).

Regulatory arbitrage feeds the expertise chasm directly. When AI systems are deployed across dozens of jurisdictions with divergent regulatory philosophies, no single regulator sees the full picture. Oversight capacity fragments across agencies and borders, and accountability becomes functionally impossible.

4.4. Expertise Chasm

The expertise chasm emerges when regulators lack the technical capacity to verify industry claims independently, creating structural dependency on the very companies they are meant to oversee. This replicates the dynamic that preceded the 2008 financial crisis, when CFTC Chairman Gary Gensler testified that regulators assumed sophisticated institutions would self-discipline through market forces, an assumption proven catastrophically wrong as "these 'sophisticated' participants had countervailing incentives to assume

risks to boost revenues" (Gensler, 2010).

The evidence that this pattern is repeating in AI governance is extensive. NIST's own March 2026 report, *NIST AI 800-4: Challenges to the Monitoring of Deployed AI Systems*, identifies "hiring and training qualified AI experts" as a systemic barrier to oversight (National Institute of Standards and Technology, 2026). That agency's capacity was simultaneously being dismantled: in February 2025, the Trump administration planned to eliminate 497 NIST positions, effectively gutting the AI Safety Institute (Gold & Curi, 2025). A January 2026 survey of federal agencies found that 55% of respondents cited a lack of employees with skills to implement AI as the primary bottleneck to AI oversight (Kelley, 2026). Median privacy and AI oversight staff sizes dropped from eight to five year-over-year across agencies (Maundrill, 2026).

The chasm is not limited to federal agencies. A March 2026 Brookings analysis of state-level AI legislation found that states with low levels of STEM workforce concentration and knowledge institutions tend not to pass AI bills, with legislators explicitly lacking "local experts to consult, so risk assessment and policy design become harder due to knowledge gaps" (Denford et al., 2026).

The expertise chasm is further compounded by AI's general-purpose nature. The gap applies not only to core AI technologies but to every application domain in which those technologies are deployed. When Anthropic reported in November 2025 that it had detected Claude being used to perform cyberattacks with 90% autonomy, the disclosure was received positively by the ML community but split the cybersecurity community (Anthropic, 2025; Gumbley, 2025). Security experts questioned why AI capabilities appear so much larger for malicious actors than for ethical hackers (Goodin, 2025), while analysts noted the report lacked standard operational security details, including the implausibility of a state actor using a proprietary, fully logged LLM (Ulili, 2025). This illustrates a combinatoric problem: technical knowledge of both frontier AI models and cybersecurity is necessary to properly set standards in that domain alone. Multiply this across every domain in which general-purpose models are deployed, and the expertise requirement exceeds what any single agency can staff.

The consequences are already measurable. As documented in Section 2.2, OpenAI's systems flagged 377 self-harm messages during conversations with Adam Raine yet never terminated sessions or alerted authorities (Raine & Raine, 2025). Without independent audit capacity, this gap is undetectable until deaths occur. The UK AI Safety Institute provides a counterexample demonstrating what independent evaluation can achieve: AISI-conducted evaluations of over 30 frontier models, including pre-deployment access to at least three, found exploitable vulnerabilities in every system tested and self-replication success rates rising from approximately 5% to over 60% between 2023 and 2025 (AI Security Institute Technical Staff, 2024).

The expertise chasm completes the self-reinforcing cycle back to regulatory capture. The Future of Life Institute's AI Safety Index states directly: "AI developers control both the design and disclosure of dangerous capability evaluations, creating inherent incentives to underreport alarming results or select lenient testing conditions that avoid costly deployment delays" (Future of Life Institute, 2025). When Sam Altman testified before the Senate, he framed OpenAI's own evaluation methodology as the baseline for what safety testing should entail (Altman, 2023). The *Illinois Journal of Law, Technology & Policy* documents the result: "AI regulation typically brings in regulated entities as joint problem-solvers with governments supposed to steer, not row" (Lévesque, 2025). When the only available expertise belongs to the industry being regulated, capture is not a risk. It is the default outcome.

4.5. How the Dynamics Reinforce Each Other

Figure 2 illustrates six causal mechanisms through which these dynamics form a self-reinforcing system.

Regulatory capture *prevents verification requirements*, directly enabling the hype-capability gap, and *blocks regulatory harmonization*, preserving the jurisdictional fragmentation that regulatory arbitrage exploits (Section 4.1).

The hype-capability gap creates *market pressure that seeks the weakest jurisdiction*, feeding regulatory arbitrage, and generates *false benchmarks that mislead oversight*, deepening the expertise chasm (Section 4.2).

Regulatory arbitrage ensures that *no single regulator sees the full picture*, widening the expertise chasm (Section 4.3).

The expertise chasm completes the cycle as *industry becomes the only expert*, and capture becomes self-sustaining (Section 4.4).

No single intervention can break this cycle. This is why the framework proposed in Section 5 requires both independent technical expertise ("Brains") and enforceable cross-border standards ("Teeth").

5. Call to Action: The "Brains and Teeth" Framework

We propose a "Brains and Teeth" framework in which the research community provides independent technical expertise ("Brains") and regulators provide enforcement authority ("Teeth") to prevent the regulatory capture dynamics documented in Section 4. This is a normative proposal for

institutional reform, not a description of current arrangements. The framework draws on proven precedents from pharmaceutical regulation, automotive safety, cryptographic standards, and cross-border banking coordination

5.1. Independent Verification of Capability Claims

We call for mandatory independent verification of AI capability and safety claims through a three-party structure separating funders, conductors, and regulators.

5.1.1. Mandatory Pre-Deployment Audits

We propose that AI systems making safety-critical claims must undergo independent audits before deployment, conducted by accredited university laboratories or nonprofit research institutions with no industry conflicts. Companies claiming safety features such as suicide prevention, harm detection, or content filtering must submit production systems for independent red-teaming, not special audit versions. Companies must submit production systems as accessed by end users for independent red-teaming, not special audit versions. Companies fund audits through deployment fees but cannot select investigators, review results before publication, or influence test design.

We recognize that revolving door incentives remain a vulnerability even with these safeguards. Mandatory cooling-off periods restricting auditors from accepting employment at companies they reviewed for a defined period after conducting reviews would provide an additional structural safeguard

This structure has direct precedent in pharmaceutical regulation. The FDA requires companies to fund clinical trials, but investigators at universities, hospitals, and academic medical centers conduct them, not company staff (U.S. FDA, 2018). Independence is maintained through conflict-of-interest prohibitions on review board members (U.S. FDA, 2025) and financial disclosure requirements (Kleppinger, 2019). NHTSA's automotive crash testing strengthens this model further: independent contractors conduct testing, and test samples are purchased through retail channels to eliminate manufacturer sample selection (Hershman, 2001).

For AI, we propose analogous safeguards. Investigators sign legally binding independence commitments with criminal penalties for falsifying safety test results. Independent AI Safety Review Boards, staffed by researchers with disclosed financial interests who are prohibited from reviewing systems where conflicts exist, oversee the audit process. The UK AI Safety Institute's independent evaluation of over 30 frontier models (Section 4.4) demonstrates that this approach is already technically feasible and reveals safety gaps that self-reporting conceals.

5.1.2. Penalties for Misrepresentation

Independent verification is meaningless without enforcement consequences. We propose penalty structures modeled on existing automotive and pharmaceutical enforcement.

Ford Motor Company paid $165 million in civil penalties in 2024 for failing to comply with federal recall requirements (NHTSA, 2024). Consent orders additionally required Ford to build testing infrastructure and review all recalls issued over three years (Lowery, 2023). The FDA enforces through criminal penalties for falsifying clinical trial data and civil penalties of $10,000 per day for continuing violations (U.S. FDA; Loeb & Loeb LLP, 2020).

For AI systems, companies deploying systems without required independent audits face civil penalties of $100K–$1M (*Suggestive figures*) per day until compliance is achieved. When safety claims fail independent verification yet companies marketed those claims to users, penalties scale to $10M (*Suggestive Figure*) per day. Falsifying safety test results or providing special audit versions that differ from production systems constitutes a federal criminal offense with imprisonment provisions.

5.2. Joint AI-Legal Institutions for Standards Development

We propose an AI Safety Standards Center with three integrated divisions. The Technical Standards Division employs machine learning researchers, AI safety researchers, formal verification experts, and domain specialists for high-risk application areas including healthcare, cybersecurity, criminal justice, and financial services who develop safety benchmarks and testing protocols through open competitions. The Legal Translation Division staffs attorneys with expertise in technology regulation and international law who translate technical standards into enforceable regulatory language. The Implementation Support Division employs regulatory scientists and policy analysts who provide guidance to government agencies on adopting and enforcing standards.

When technical experts specify that AI systems must pass adversarial testing for suicide prevention, legal experts define what "pass" means in enforceable terms, establish burden of proof requirements, specify remedies for failures, and draft regulatory text that courts can interpret consistently.

This structure has precedent in NIST's cryptographic standards development, where government institutions successfully leveraged academic expertise through open competitions, public review periods, and formal researcher participation while maintaining independence from industry influence (Regenscheid, 2016).

5.3. Cross-Border Technical Standards

We call for cross-border technical standards that define verification protocols enabling intergovernmental coordination without requiring legal harmonization. As documented in Section 4.3, regulatory arbitrage is actively engineered through lobbying that blocks harmonization and then exploits the resulting fragmentation.

We propose core technical protocols that define standard formats for reporting AI system capabilities, training data characteristics, safety testing results, and deployment contexts. Home country regulators, where AI companies are headquartered, oversee consolidated safety assessments covering all deployments globally. Host country regulators, where AI systems are deployed, verify that systems meet local requirements for high-risk applications such as systems affecting children, mental health services, criminal justice, or financial decisions. Technical protocols specify data interchange formats enabling automatic verification that a system approved in one jurisdiction meets another jurisdiction's requirements when those requirements reference the same underlying technical standards.

This structure draws on the Basel Committee on Banking Supervision, which achieved cross-border coordination through technical standards combined with institutional design dividing responsibilities between home country supervisors responsible for consolidated oversight and host country supervisors responsible for local entities (Basel Committee on Banking Supervision, 1996). The Committee established principles for dividing responsibilities while avoiding redundant validation work (Basel Committee on Banking Supervision, 1996). For AI, this means that if both the EU and the US require suicide prevention testing against independent benchmarks, and a system has passed EU-approved testing, the protocol enables recognition of that testing for US deployment approval, avoiding redundant work while maintaining independent oversight.

6. Alternative Views

6.1. Not All Big Tech Actors Act in Bad Faith

A reasonable objection is that not all Big Tech actors act in bad faith, with some companies investing substantially in safety research (Shah et al., 2025; Fronsdal et al., 2025) and publishing findings openly (Lindsey, 2025; Marks et al., 2025; Lynch et al., 2025; McCain et al., 2025; Cunningham et al., 2026). This observation is accurate. However, regulatory frameworks cannot depend on good faith alone. The "Brains and Teeth" framework we propose benefits companies acting in good faith by creating clear, verifiable standards that reward genuine safety investments and enable differentiation from competitors making inflated claims. The framework must work even when actors prioritize profit over safety, because that is precisely when accountability mechanisms are most essential.

6.2. First Principles Might Be Better Than Historical Analogy

This view argues that first principles thinking might be superior to historical analogy given AI's unique characteristics. As documented in Section 3.2, AI presents challenges fundamentally different from previous technologies. First principles analysis of these properties is valuable, but it does not require ignoring historical patterns of regulatory capture. We learn from finance and fossil fuels not because AI is identical to banking or oil extraction, but because the capture dynamics are structural, as documented in Section 2.1, and repeat across industries regardless of the underlying technology. First principles and historical learning are not mutually exclusive. The question is whether we learn from documented regulatory failures and proven accountability mechanisms, or whether we allow AI governance to repeat patterns that produced the 2008 financial crisis and decades of climate science suppression.

7. Conclusion

Accountability frameworks take years to develop. By the time institutions mature and standards become effective, deployment patterns are entrenched and lobbying relationships are established. We do not have decades to get this right. The AI research community has comparative advantages that make leadership both possible and necessary: proximity to the frontier enables keeping pace with rapid development, open science traditions create transparency that proprietary industry cannot match, and institutional independence from profit motives allows prioritizing safety over market pressure. History demonstrates that large industries cannot be trusted to self-regulate when profits conflict with public interest, and AI is following the identical pattern. The question is whether those frameworks will be established through independent research community leadership grounded in technical expertise and open science, or through industry capture that writes rules protecting profits rather than people. The research community must lead now. The alternative is regulatory capture defining AI governance for the foreseeable future.

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